



Samsung Innovation Campus

Graduation Project

ATHARNA

Smart Museum Security & Visitor Experience System

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Abstract

This project introduces **ATHARNA**, a smart IoT-based system designed to proactively protect museum artifacts and significantly enhance visitor experiences. It integrates multiple physical and environmental sensors (**Load Cell, Ultrasonic, Gas, and DHT sensors**), multi-layered authentication (**NFC and Facial Recognition**), and a modern mobile application. The entire system is unified via the **MQTT protocol (HiveMQ broker)**, enabling real-time data visualization on a **Node-RED dashboard** and a **Flutter mobile app**. This integrated approach provides continuous environmental monitoring, robust access control, and visitor management, marking an innovative step toward improved artifact safety and operational efficiency in modern museums.

1. Introduction

Museums represent invaluable cultural and historical institutions that necessitate advanced security and environmental monitoring solutions. Traditional surveillance and manual monitoring methods are frequently insufficient to prevent theft, unauthorized handling of artifacts, or subtle environmental degradation.

The **Smart Museum Security & Visitor Experience System (ATHARNA)** aims to combine modern IoT technology, computer vision, and mobile applications to offer a comprehensive and intelligent protection solution that addresses the shortcomings of conventional security setups.

2. Problem Definition

Current security practices in many museums rely on outdated human supervision and simple alarms, which are prone to human error and often result in delayed response times in critical situations (such as theft or fire). Moreover, there is often a lack of integration between environmental monitoring systems and physical access control mechanisms, which can lead to inefficient operations and increased risk of loss or damage of artifacts.

3. Objectives

The primary objectives of the ATHARNA project are:

- **Enhance Artifact Security:** Implement multi-sensor protection (weight and proximity) to detect unauthorized handling or removal in real-time.
- **Secure Access Control:** Utilize **Facial Recognition (OpenCV)** and **NFC authentication** to ensure only verified personnel (e.g., restoration staff) can approach or handle artifacts.
- **Environmental Safety:** Continuously monitor and log critical environmental parameters, including **temperature, humidity (DHT11)**, and potential **gas/fire hazards (MQ2)**.
- **Real-Time Data Access:** Provide immediate alerts and comprehensive data visualization through both a web-based dashboard (**Node-RED**) and a cross-platform mobile application (**Flutter**).
- **Improve Visitor Experience:** Offer visitors a streamlined experience through mobile booking, reservation management, and interactive artifact information via **QR code scanning**.

4. Literature Review

Recent technological advancements in the Internet of Things (IoT) and smart surveillance systems have demonstrated significant potential for asset protection across various sectors. Studies highlight the effectiveness of combining various sensor types and computer vision techniques for real-time threat detection. However, few existing systems integrate comprehensive artifact physical security, environmental preservation, and sophisticated visitor management into a single, cohesive platform. This project is specifically designed to bridge this gap by offering a hybrid, all-in-one approach.

5. System Architecture

The proposed system operates on a decentralized architecture, utilizing the **Raspberry Pi** as the central edge processing unit. Data communication across the network is standardized using the lightweight **MQTT protocol** with the **HiveMQ** broker for reliability.

Key Components:

Component	Category	Purpose
Raspberry Pi	Edge Controller	Main processing unit for sensor reading, Python logic, and OpenCV execution.
Weight Sensor (Load Cell)	Security	Detects artifact displacement or removal; crucial anti-theft trigger.
4x Ultrasonic Sensors	Security	Monitors proximity; creates a virtual protective barrier around the exhibit.
Gas Sensor (MQ2)	Environmental	Detects harmful gases and smoke/fire hazards; triggers physical alarm.
DHT11 Sensor	Environmental	Measures and transmits ambient temperature and humidity levels.
NFC Module	Access Control	Authorizes visitor and staff entry/exit, contributing to accurate visitor count.
Camera Module	Computer Vision	Captures images for real-time facial recognition and staff verification.
MQTT (HiveMQ)	Communication	Ensures reliable, low-latency, real-time data exchange.
Node-RED Dashboard	Application	Visualization of all sensor data, visitor counts, and security alerts.
Flutter App	Application	Mobile interface for booking, cancellations, and QR-based artifact details.

6. Methodology

6.1. Hardware Implementation

The hardware layer is designed for continuous, multi-parameter monitoring.

- **Artifact Integrity:** Each protected artifact is secured by both a **Load Cell sensor** (detecting lift or pressure changes) and **four Ultrasonic sensors** (detecting unauthorized close proximity).
- **Environmental Monitoring:** The **DHT11** and **MQ2** sensors are constantly sampling air quality. If the MQ2 detects a threat, a buzzer is immediately triggered via the Raspberry Pi.
- **Access Control Flow:** The **NFC reader** grants entry/exit, while the **Camera Module** runs **OpenCV** to verify staff identity against an authorized database. Unrecognized faces or unauthorized NFC tags trigger security protocols.

6.2. Software Implementation

The software is structured into three distinct layers:

1. **Edge Layer (Raspberry Pi):** Python scripts handle sensor polling, signal processing, and the execution of the Face Recognition model. Cleaned data is then published to the Cloud Layer via MQTT.
2. **Cloud Layer (HiveMQ Broker + Firebase):** The MQTT broker facilitates all real-time communication. Firebase is used specifically to manage the Flutter app's data (user accounts, booking records).
3. **Application Layer (Node-RED + Flutter):** Node-RED subscribes to MQTT topics to display security and environmental data instantly. The Flutter app uses Firebase for transactional data (booking) and may interact with the MQTT broker for real-time admin notifications.

6.3. Data Flow

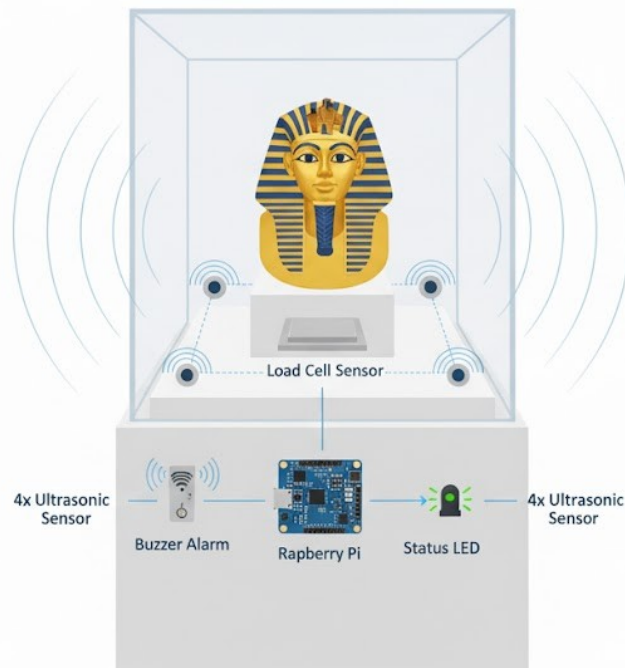
1. **Sensors** collect data and transmit to the **Raspberry Pi**.
2. The **Pi** processes sensor data, executes **Facial Recognition**, and publishes all readings to the **HiveMQ Broker** via **MQTT**.
3. The **HiveMQ Broker** broadcasts the data to all subscribers: **Node-RED** and the **Flutter App**.
4. **Node-RED** displays real-time readings, status, and logs security alerts.
5. The **Flutter App** handles user authentication and booking (via Firebase) and provides mobile monitoring for administrators.

7. Results

The ATHARNA system achieved successful integration and functionality across all key modules:

- **Security Validation:** The combination of Load Cell and Ultrasonic sensors successfully detects both artifact removal and proximity violation, proving the double-layered security concept.
- **Real-time Monitoring:** The system accurately reads environmental parameters, with smooth, real-time updates displayed on both the Node-RED dashboard and the mobile application.
- **Authentication:** The dual-factor security using NFC and Face Recognition effectively restricts access to authorized personnel.
- **Visitor Management:** The Flutter application provides a robust and user-friendly interface for booking, managing reservations, and accessing QR-based artifact information.

Physical Artifact Security Setup



8. Future Work

To further advance the ATHARNA system, the following enhancements are proposed:

- Integration with **AI-based Threat Prediction Models** for analyzing visitor movement patterns to detect suspicious behavior before an incident occurs.
- Adding **GPS-based tracking** capabilities for mobile artifacts or staff assets.
- Expanding the database and optimizing the model for higher **Facial Recognition accuracy** and speed.
- Integrating with existing museum Enterprise Resource Planning (ERP) systems for full operational automation.

9. Conclusion

The **Smart Museum Security & Visitor Experience System (ATHARNA)** successfully meets its objectives by effectively combining IoT sensors, computer vision, and mobile technology to protect valuable museum artifacts. Through multi-layered authentication, continuous environmental monitoring, and intelligent real-time alerts, the system represents an innovative, practical, and highly scalable solution for creating smarter and safer museums.

10. References

- Internet of Things (IoT) Architecture for Smart Security Systems.
- Research Papers on Raspberry Pi-Based Surveillance and Access Control.
- MQTT Communication and HiveMQ Documentation.
- Node-RED and Flutter Official Documentation.